



Enhancing the carotenoid recovery from *Eugenia uniflora* L. pulp: A comparative study of supramolecular solvents, ionic liquids, and deep eutectic solvents

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ABSTRACT

Carotenoids are highly valuable compounds for food, pharmaceutical, and cosmetic applications, given their natural biological activities. Due to their hydrophobic nature, the processes of separation and purification present some limitations. This work aimed to promote a comparative study on different classes of green solvents: deep eutectic solvents (DES), ionic liquids (ILs), and supramolecular solvents (SUPRAS). The solvents were compared regarding the ability of each solvent to recover natural carotenoids from *Eugenia uniflora* L. pulp and the preservative effect of the solvent against thermal and photodegradation. The solvent recycling and its capacity to be reused were also evaluated. The results indicate that SUPRAS ($60.45 \pm 0.55\%$) promoted a higher recovery yield than ILs ($29.72 \pm 0.27\%$) and DES ($9.50 \pm 0.23\%$). Besides, the higher thermal stability of the carotenoids was observed for the carotenoids-rich extracts obtained with DES (at $40\text{ }^{\circ}\text{C}$, half-life time of 353.39 h and $50\text{ }^{\circ}\text{C}$ half-life time of 279.44 h) and SUPRAS (at $60\text{ }^{\circ}\text{C}$, half-life time of 143.13 h). The thermodynamic parameters showed superior activation energy for AC: ET and DES-based extracts, besides demonstrating that the degradation processes had positive free Gibbs energy and enthalpy and negative entropy values. For the light stability assays, IL had the highest protective capacity against UV (half-life time 54.6 h) and LED (half-life time 216.6 h) lights. Finally, the best systems were integrated with a liquid–liquid system envisioning carotenoid recovery, solvent recycling, and economic viability evaluation. In this context, the SUPRAS could be reused for up to four cycles compared to three cycles of ILs, besides providing a lower process cost than ILs solutions.

1. Introduction

Eugenia uniflora L., also known as Pitanga, is a species of the Myrtaceae family, commonly found in Brazil. Owing to its adaptability, this species is found in a wide area from the north to the south of Brazil, especially in the Atlantic rainforest [1]. The pitanga fruits are composed of approximately 77% pulp and 23% seed, wherein its pulp is rich in minerals (e.g., potassium and calcium) and bioactive compounds (e.g., carotenoids, flavonoids, vitamin C, and anthocyanins) [2]. Among the main carotenoids, the red pitanga is predominantly rich in β -carotene and lycopene [3]. Therefore, the extraction of bioactive compounds from pitanga pulp (EUP) throughout extraction processes has been investigated due to EUP health benefits, such as antioxidant, anti-

inflammatory, and antidiabetic properties [4,5].

Typically, the extraction of carotenoids from plant matrices exhibits some limitations regarding the hydrophobic nature of these compounds and their susceptibility to excessive exposure to heat, light, acids, oxygen and long extraction times [6]. Therefore, carotenoids are conventionally obtained with non-polar solvents (e.g., hexane, petroleum ether, acetone, and tetrahydrofuran) that display compatible similarity with the target compounds. However, most of these solvents present several drawbacks as regards their toxicity, environmental issues, and human health risks. As a result of this, eco-friendly and designer solvents such as ionic liquids (ILs), deep eutectic solvents (DES), and supramolecular solvents (SUPRAS) have been proposed as efficient replacements for the conventional solvents to extract lipophilic bioactive compounds.

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ILs are salts commonly composed of organic cations and inorganic/organic anions. Depending on their composition, ILs can be classified as sustainable and cost-effective solvents to recover bioactive compounds from a wide range of plant matrices [7,8]. Regarding the ILs-forming compounds, cholinium cation ($[Ch]^+$) in combination with carboxylic acids has been proven to be a promising alternative due to its biocompatible and biodegradable properties, besides being recyclable and reusable in a new SLE process [8,9]. In addition to the ILs, DES are solvents composed of a hydrogen donor (HBD) and an acceptor (HBA) molecule, which can also be a sustainable solution since some mixtures are biodegradable and non-toxic. In this sense, DES formed by DL-Menthol and carboxylic acids has been applied in SLE to present these features, in addition to its capacity to efficiently extract lipophilic bioactive compounds from plant matrices [10]. Besides ILs and DES, SUPRAS has been used in the SLE of solutes for being a sustainable and tailor-made solvent [11]. Hence, SUPRAS are described as nanostructured liquids composed of amphiphilic molecules (e.g., carboxylic acids) formed by the self-aggregation and the coacervation processes, in which the target solutes can be solubilized and entrapped in the supramolecular microenvironment due to the several binding sites formation in the solvent [11,12].

The employment of these alternative solvents in SLE is of great interest to the industry envisioning the development of a biocompatible and environmental-friendly platform. In this sense, another factor to be considered is the recyclability of the solvent. This strategy makes it possible to reduce the amount of solvent required, the high time consumption and the low recovery yields, besides following the circular economy precepts. Therefore, this work aims to evaluate three classes of green solvents (ILs, DES, and SUPRAS) to extract and recover carotenoids from EUP. Hence, ILs, DES, and SUPRAS composed of octanoic and decanoic acid were evaluated regarding their extraction aptitude by the total carotenoid content (TCC) analysis and considering their capacity to preserve the carotenoids throughout thermal, UV-light, and LED light exposure. Moreover, the mechanism of carotenoid recovery from EUP samples was also investigated by scanning electron microscopy analysis. Lastly, the feasibility of recovering the carotenoid compounds and reusing the solvents based on an integrated platform was investigated.

2. Material and methods

2.1. Chemicals

DL-Menthol (purity $\geq 95\%$), choline bicarbonate (purity $\sim 80\%$), octanoic acid (purity $\geq 99\%$), decanoic acid (purity $\geq 98\%$), lycopene (purity $\geq 98\%$), β -Carotene (purity $\geq 93\%$), Ethyl acetate anhydrous (purity $\geq 99.8\%$) were purchased from Sigma-Aldrich. Hydrochloric acid (purity $\geq 37\%$) was purchased from Anidrol, and ethanol (purity $\geq 99.5\%$) from Panreac. Acetone (purity $\geq 99\%$) and hexane (purity $\geq 99\%$) were purchased from Neon and Exôdo Científica, respectively. All experiments used ultrapure water from Milli-Q® Direct Water Purification System.

2.2. Obtaining and preparation of *Eugenia uniflora* L. Pulp (EUP)

The EUP fruits were collected at the Canguiri experimental farm (Federal University of Parana) (SISGEN authorization AA5E97; Exsiccate voucher n° MBM068565). The samples were manually pulped and lyophilized (L101 - Liotop® Brazil). Then, the freeze-dried fruits were properly ground, vacuum packed, and stored at -18°C in polyethylene bags.

2.3. DES, ILs, and SUPRAS preparation and extraction

The DES preparation was performed by weighting the forming compounds (HBA and HBD) in an analytical balance (AUX320, Shimadzu, Japan), followed by mixing the components at the molar ratio of

1:1. Sequentially, the mixture was heated in a Dubnoff bath (Ethik Technology) at 50°C , under constant agitation until the formation of a homogeneous solution [13]. The ILs preparation was synthesized as described by Mussagy and co-authors in 2019 and 2020 via acid-base neutralization [14,15]. Thus, the reaction between the compounds was stirred at room temperature for 72 h, followed by the purification in a rotary evaporator (MA120/THV – Marconi) at 60°C under a vacuum (700 mmHg). After that, DES and ILs were dissolved in ethanol (50 %, w/w), and the mixtures were employed in SLE. The DES and ILs constituents are shown in Fig. 1.

The SLE of EUP carotenoid compounds using DES and ILs solutions were carried out in a jacketed glass cell coupled to a thermostatic bath (RW-1025G, Lab Companion) under constant magnetic agitation (1500 rpm). The extraction conditions such as time and solid-liquid ratio extraction were set at 1 h and 1:40 (w/v), respectively. The effect of temperature on SLE efficiency was studied at 25 and 45°C .

The SUPRAS was produced simultaneously with the SLE procedure. Therefore, the amphiphilic molecule (5.0 %, v/v) was dissolved in ethanol (27.5 %, v/v), followed by the addition of the EUP sample (1:40 w/v) and acidified water (pH ~ 3) (67.5 %, v/v) (Fig. 2). Subsequently, the mixture was shaken on a vortex mixer (Gomixer, MX-S) for 3 min, at 25°C , and centrifuged at 1200 r/min for 15 min (Thermo Fisher Scientific Fresco 21 Micro Centrifuge) to separate the phases.

Carotenoid extraction from the EUP samples was also evaluated using a mixture of acetone and ethanol 1:1 (v/v) (AC:ET) as a comparative solvent. The SLE was performed at 25°C for 1 h at a ratio of 1:40 (w/v) and under constant magnetic stirring (1500 rpm). The extraction yield was calculated by determining the initial carotenoid content in the EUP samples. For this purpose, acetone and hexane (7:3) (v/v) were mixed with the EUP sample (1:40 (w/v)) and shaken at 25°C (Ethics Technology). This procedure was repeated for three cycles until the EUP was colorless.

2.4. Evaluation of carotenoids on EUP-based extracts

The spectrophotometric measurement of total carotenoid content (TCC) was conducted using the UV/Vis Spectrophotometer (UV-1800 Shimadzu), with a 1 cm optical path-length quartz cell, at a wavelength of 450 nm, on the EUP samples. To determine the concentration of TCC in each solvent, the Beer-Lambert law was employed, and a calibration curve was constructed using a β -carotene standard. The calibration curve displayed a linear relationship between absorbance and carotenoid concentration, thereby relying on the principles of the Beer-Lambert law. Finally, the yield of total carotenoids ($\mu\text{g/g}$) in each solvent was calculated using Eq (1). [16]:

$$TCC = \frac{A \times V \times 10^4}{A_{1\text{cm}}^{1\%} \times m} \quad (1)$$

Where: TCC is the total concentration of carotenoids ($\mu\text{g/g}$); A is the absorbance at 450 nm; V is the volume of solvent (mL); m is the EUP mass (g), and $A_{1\text{cm}}^{1\%}$ is specific molar absorptivity.

2.5. Stability of total carotenoids

The stability of EUP-based extracts obtained with DES and ILs at 45°C and with SUPRAS phase and AC:ET at 25°C was evaluated based on thermal degradation and light exposure. The effect of the temperature was analyzed using an Eppendorf® ThermoMixer® at 40°C , 50°C , and 60°C in the absence of light. Total carotenoid losses were assessed for 3 h at 0.5-h intervals and calculated as described previously in this section (Eq. (1)).

Subsequently, thermal stability was calculated by determining the reaction rate constant (k) (Eq. (2)) and the half-life time ($t_{1/2}$) (Eq. (3)) using a first-order reaction model for 50% of TCC degradation.

$$\ln(C/C_0) = -kt \quad (2)$$

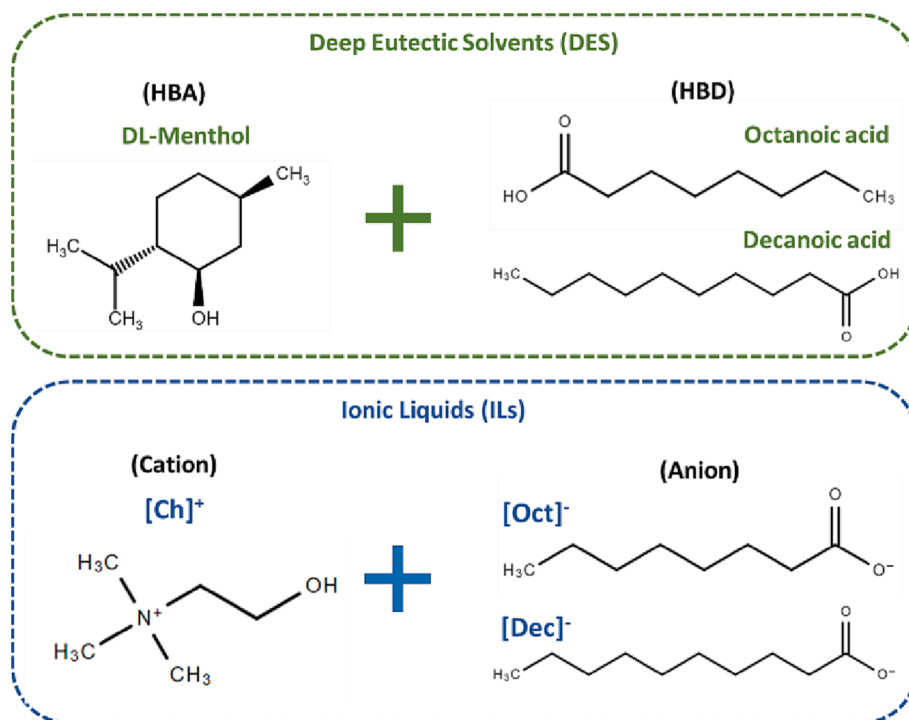


Fig. 1. Chemical structures of DES (HBA and HBD) ILs (cations and anions) precursors employed in this work.

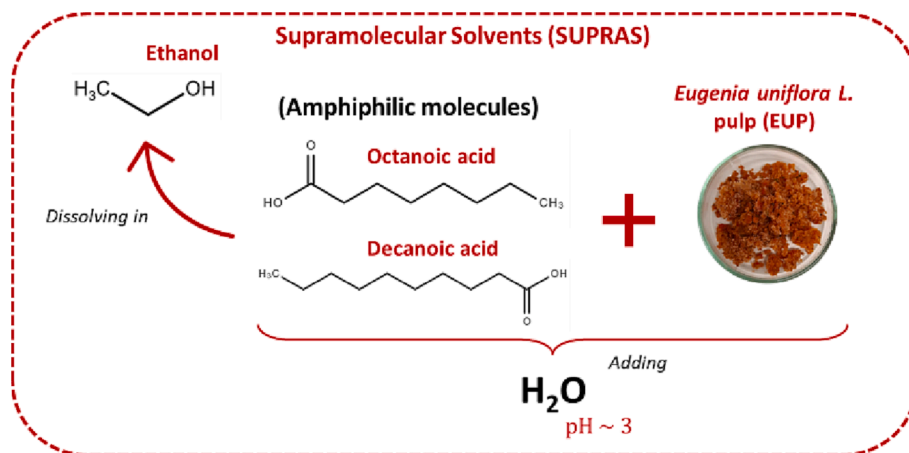


Fig. 2. Illustration of SUPRAS-based simultaneous production and SLE of EUP carotenoids compounds.

$$t_{(1/2)} = -\frac{\ln(0.5)}{k} \quad (3)$$

where t is the time of storage (days), C_0 and C are the initial and final concentrations of total carotenoid content, respectively.

In the sequence, the effect of temperature on thermal degradation was evaluated using the Arrhenius equation (logarithmic form) (Eq. (4))

$$\ln k = -\frac{E_a}{RT} + \ln A \quad (4)$$

where k is the reaction constant at the reference temperature (T) in Kelvin, E_a is the activation energy (kJ/mol), R is the universal gas constant (8.314 J/mol. K) and A is the preexponential factor (h^{-1}). The E_a value was obtained by linear regression analysis.

Moreover, the activation enthalpy (ΔH) and entropy activation (ΔS) were determined according to the activated complex theory (Eq. (5)).

$$\ln\left(\frac{k}{T}\right) = -\left(\frac{\Delta H}{RT}\right) + \left(\ln\frac{k_b}{E_i}\right) + \left(\frac{\Delta S}{R}\right) \quad (5)$$

where k is the reaction constant at reference temperature (T) in Kelvin, k_b is the Boltzmann constant (1.3806×10^{-23} J/K), and E_i is the Planck constant (6.6260×10^{-34} J.s). The ΔH and ΔS value were obtained by linear regression analysis.

Lastly, the Gibbs energy of activation (ΔG) was determined by the relationship between ΔH , ΔS , and degradation temperature (T) (Eq. (6)).

$$\Delta G = \Delta H - T\Delta S \quad (6)$$

The light stability of EUP-based extracts was evaluated at room temperature (25.0 ± 0.2 °C) by exposing the samples to an LED lamp (A67, 15 W, 30 cm distance) and a UV-light (9 W, UV lamp, 30 cm distance). The carotenoid degradation was examined at 0.5-hour intervals for 3 h. In sequence, the reaction rate constant and half-life time were calculated using (Eq. (2) and Eq. (3)).

2.6. Solvent recovery

The recovery of ILs solution was performed by applying a back-extraction procedure employing ethyl acetate as the antisolvent. Therefore, IL-based EUP extracts (40% v/v) were mixed with water (30% v/v) and pure ethyl acetate (30% v/v) to produce a biphasic system (BS). The BS was centrifuged (Thermo Fisher Scientific Fresco 21 Micro Centrifuge) for 15 min at 1200 rpm and 25 °C for phase separation. Thus, the top-rich carotenoid phase (*i.e.*, the phase consisting of ethyl acetate) was analyzed through the TCC quantification. Concerning the ILs ethanolic solution (*i.e.*, the bottom phase), the ethanol amount was removed using a rotary evaporator (MA120/THV – Marconi) at 50 °C for 4 h. The ILs were reused in two consecutive cycles to extract the carotenoids from the EUP samples.

The recovery of SUPRAS phase was accomplished at room temperature by adding excess water to the SUPRAS-based EUP extracts to separate the amphiphilic molecule (*i.e.*, octanoic acid or decanoic acid). Subsequently, the amphiphilic rich phase was mixed with an aqueous solution of NaOH 1 M to adjust the pH. After changing the pH > 7, ethyl acetate was added to the mixture for the BS formation (ethyl acetate top-rich phase and sodium octanoate or sodium decanoate bottom-rich phase) and centrifuged at 1200 rpm at 25 °C for 5 min. The top-rich carotenoid phase was evaluated in terms of TCC. Finally, recycling of the amphiphilic molecules was performed by adding an aqueous solution of HCl 1 M to adjust the pH (pH < 7). Thereafter, the octanoic acid was reused in three consecutive cycles in the SUPRAS simultaneous production and SLE of EUP-carotenoid.

Finally, the ethyl acetate employed to recuperate the carotenoids from the ILs solution and SUPRAS-based extracts was also recovered using a rotary evaporator for 30 min at 40 °C and reused for further extraction cycles. The recovery yields (%) of carotenoids from EUP-based extracts were calculated as the ratio between the initial concentration of TCC (C_i) in the samples and the final concentration (C_f) after the solvent recovery process (Eq. (7)).

$$\text{Recovery of TCC (\%)} = \frac{C_i (\mu\text{g/g})}{C_f (\mu\text{g/g})} \cdot 100 \quad (7)$$

2.7. Statistical analysis

The TCC results were statistically analyzed using Statistic 10.0 software (StatSoft USA). The Shapiro-Wilk test ($p > 0.05$) verified the normality hypothesis, and the Bartlett test confirmed homoscedasticity. In sequence, analysis of variance (ANOVA) ($p < 0.05$) was then performed using Tukey's HSD test to evaluate mean differences.

3. Results and discussion

3.1. Carotenoids extraction from EUP

The extraction of carotenoids from EUP pulp was evaluated by employing different solvents: ILs, DES, SUPRAS phase, and AC:ET. SUPRAS-based extraction is performed simultaneously with solvent formation, using vortex mixing for 3 min. Taken this into consideration, a previous screening of the SLE conditions was conducted at 25 °C for 3 min also for all solvents to investigate whether an increase in the extraction time and temperature was required. The results are shown in Fig. 3.

As shown in Fig. 3, the carotenoid recovery yield promoted by ILs, DES, and AC:ET for 3 min at 25 °C was <13%. Nonetheless, a yield of over 58% was obtained with the SUPRAS systems, suggesting that a short extraction time is required to achieve an adequate extraction capacity. This finding can be related to the supersaturation of the SUPRAS since the solvent production and the SLE process are performed in a one-step approach. As a result, the carotenoid compounds from EUP can be simultaneously solubilized in the supramolecular nanostructures and

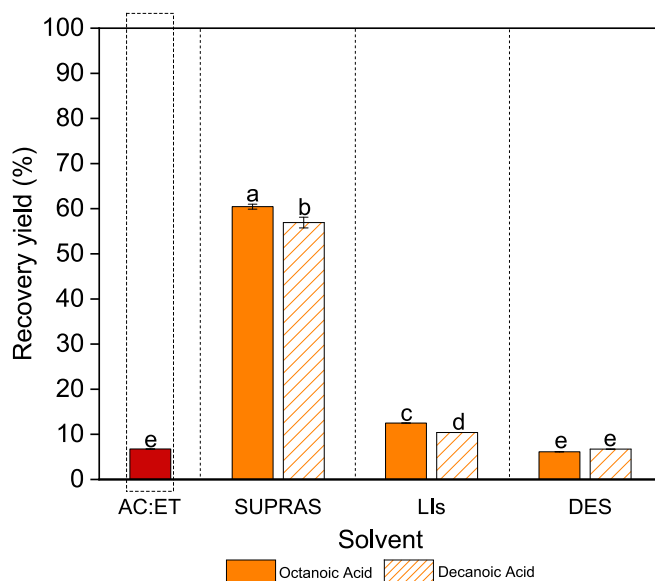


Fig. 3. Recovery yield (%) of EUP samples obtained with ILs, DES, SUPRAS phase, and AC:ET as solvents, at 25 °C for 3 min. The 1:40 (w/v) solid-liquid ratio remained for all experiments. *The values were obtained from three replicates $\pm$ standard deviation (SD). Different superscript letter for the same color of column means significant difference according to ANOVA and Tukey test ($p < 0.05$). AC:ET, mixture of acetone and ethanol 1:1 (w/w).

concentrated in the solvent. In addition, as demonstrated in previous work, increasing the vortex time does not significantly increase the recovery yield [17]. Besides, for the SLE carried out with ILs and DES, the enhancement of the temperature and the time contact between solvent and solute can boost the carotenoid recovery by improving the solute-solvent diffusivity and the mass transfer process, which facilitates the solubilization of carotenoids in the media.

Therefore, for the extraction procedures based on ILs and DES, the effect of increasing the temperature (25 °C and 45 °C) was investigated. Moreover, both solvents were dissolved in ethanol (50 %, w/w) to reduce the viscosity. For SUPRAS phase-based SLE, the following condition was established: vortex-assisted extraction at 25 °C for 3 min. In addition, the solid-liquid ratio of 1:40 (w/v) was maintained for all SLE procedures. The extraction results concerning the recovery yields are displayed in Fig. 4.

According to Fig. 4, SUPRAS phase-based extractions allowed to boost the extraction capacity of TCC from EUP samples compared to the ILs, DES, and AC:ET as solvents. The SUPRAS phase-based EUP extraction with octanoic acid as the amphiphilic molecule made it possible to recover 60.5 % of TCC. This outcome suggests that the SUPRAS phase aggregates formed through the self-assembly and coacervation processes enabled the bioactive compounds' entrapment into their inner cavity. Thus, the supramolecular network structure can offer a suitable micro-environment to extract hydrophobic compounds owing to the dispersive, dipole-dipole, and dipole-induced dipole interactions that drive the SLE process [12]. Another advantageous aspect of using the SUPRAS phase in SLE is the short duration of the process (3 min), the low temperature (25 °C), and the pH of the solvent. The SUPRAS phases, composed of octanoic or decanoic acid, exhibited the lowest pH values compared to the other solvents evaluated in this work (data available in [Supplementary Material](#), cf. [Table S1](#)). As previously reported, an acidic environment boosts the carotenoids extraction due to enhancing the degradation of the matrix cell wall [18,19].

In addition, increasing the alkyl chain length of the amphiphilic molecule did not positively affect SLE. Therefore, SUPRAS systems comprised of octanoic acid provided the highest extraction yield of TCC compared with SUPRAS systems with decanoic acid. This finding was

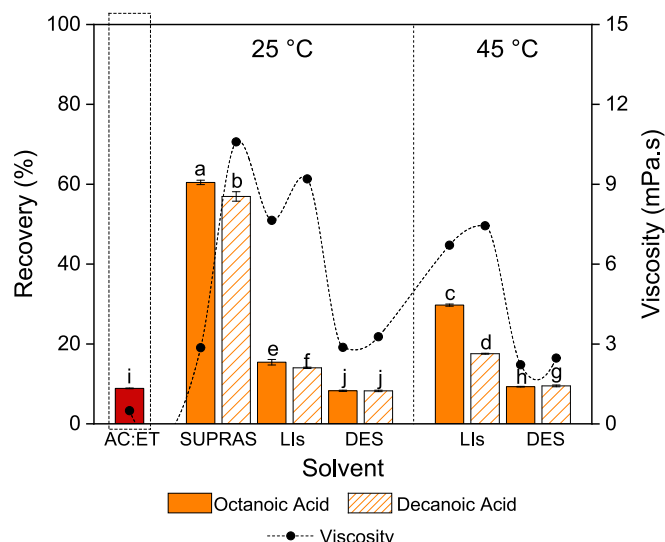


Fig. 4. Recovery yield (%) of EUP samples obtained with ILs, DES, SUPRAS phase, and AC:ET, and the solvents' viscosity (mPa). ILs and DES solutions-based experiments were carried out at 25 °C and 45 °C for 30 min. AC:ET and SUPRAS phase-based extractions were conducted at 25 °C for 60 and 3 min, respectively. The 1:40 (w/v) solid-liquid ratio remained for all experiments. *The values were obtained from three replicates $\pm$ standard deviation (SD). Different superscript letter for the same color of column means significant difference according to ANOVA and Tukey test ($p < 0.05$). AC:ET, mixture of acetone and ethanol 1:1 (w/w).

also observed for the SLE carried out with the ILs at 45 °C, indicating that increasing the degree of hydrophobicity reduces the extraction performance. Moreover, these results can be related to the solvents' viscosity (cf. Table S1). As depicted in Fig. 4, the SUPRAS phase and ILs produced with octanoic acid exhibited lower viscosity than those composed of decanoic acid. Hence, a less viscous solvent will boost the mass transfer process and the carotenoid recovery from the EUP samples. Concerning the DES-based SLE, the alkyl chain length did not significantly affect the efficiency of carotenoid extraction from the EUP samples.

Regarding ILs solutions extraction performance, the temperature rises positively influenced the SLE, particularly for the ILs formed by [Oct]-. The SLE carried out at 25 °C allowed a TCC recovery of 15.4 %, whereas the SLE at 45 °C recovered 29.7 % of the TCC. In addition, as exhibited in Fig. 4, increasing the process temperature led to a decrease in the viscosity of the ILs, which favored the mass transfer process by enhancing the solute-solvent diffusivity and the compounds' migration from the cell wall of the matrix into the solvent [20]. The same behavior was also observed for the DES-based SLE. Thus, alternative approaches to reduce the solvent viscosity, as well as employed in this work (i.e., by mixing the ILs and DES with a polar solvent), need to be considered.

Related to the extraction with AC:ET as the solvent, a higher extraction yield of TCC was found compared to the DES-based extracts obtained at 25 °C. Nonetheless, when the process temperature was increased, the DES-based SLE was more effective than the AC:ET mixture. Furthermore, the influence of the solvents in the EUP cell wall was also investigated, as shown in Supplementary Material cf. Figure S1, as well as the experimental protocol. The images were acquiring using field emission gun-scanning electron microscopy (FEG-SEM). According to Figure S1 – A (from Supplementary Material), no rupture in the cell structure of the EUP sample was observed. Concerning the images obtained after the SLE process (Figure S1 B – H from Supplementary Material), the cell wall of EUP was severely damaged. As a result, the cell structure rupture enabled the diffusivity of the solvent into the EUP sample. Regarding the SLE conducted with the SUPRAS phase and IL composed of octanoic acid, the cell structure was more damaged than

the SLE carried out with the other solvents evaluated. This outcome agrees with the SLE (Fig. 4), in which the carotenoid recovery was boosted by employing the same solvents.

3.2. Stability of EUP carotenoid compounds

3.2.1. Thermal degradation

Carotenoids are bioactive compounds highly susceptible to degradation when exposed to high temperatures, light, and oxygen [21]. Due to their highly unsaturated nature, carotenoids are susceptible to oxidation, and thermal processing can lead to their isomerization, which directly affects their bioavailability and antioxidant capacity [22]. To investigate the effectiveness of various solvents (ILs, DES, SUPRAS, and AC:ET mixture) in preserving carotenoids from EUP, a kinetics study was performed using a first-order model to fit the experimental data at 40 °C, 50 °C, and 60 °C for 3 h. The thermal stability and the thermodynamic parameters (i.e., E_a , $\Delta H^\ddagger$, $\Delta S^\ddagger$, and $\Delta G^\ddagger$) are presented in Fig. 5 and Table 1, respectively. The coefficients of determination (R^2), half-life time ($t_{1/2}$), and k values are detailed in the Supplementary Material cf. Table S2.

According to Fig. 5, EUP extracts obtained with the SUPRAS phase and DES showed higher thermal stability as a function of degradation time. Furthermore, for the assay carried out at 40 °C ($t_{1/2} = 353.39$ h) and 50 °C ($t_{1/2} = 279.44$ h), the DES composed of octanoic acid provided the highest thermal stability compared to the other solvents. Nonetheless, at a higher temperature (60 °C), the octanoic acid-based SUPRAS phase was the most suitable solvent to preserve the carotenoids from the EUP samples ($t_{1/2} = 143.13$ h) (data available in Supplementary Material, cf. Table S2). These outcomes suggests that, despite the SUPRAS_{Oct} does not exhibit the highest thermal stability at 40 °C and 50 °C, the $t_{1/2}$ of this solvent exhibited an inferior decay compared to the other solvents as the degradation temperature was increased.

In addition, the SLE carried out with ILs provided the lowest capacity to preserve the carotenoid compounds from EUP samples compared to the other alternative solvents evaluated. Nonetheless, a slight decrease in the $t_{1/2}$ with temperature enhancement was also observed. This outcome indicated that the preservation of carotenoids from EUP samples obtained with ILs was less affected by the degradation temperature. Moreover, only the mixture composed of AC:ET resulted in lower $t_{1/2}$ than the ILs (data available in the Supplementary Material, cf. Table S2).

Regarding the thermodynamic parameters (Table 1), it was noticed that among the solvents evaluated, the strongest activation energy was observed for the AC:ET (63.75 kJ/mol) and the DES-based extracts (DES_{Oct} = 45.92 kJ/mol and DES_{Dec} = 45.24 kJ/mol). This result indicates that these solvents exhibited a higher thermal sensitivity than the others tested. Hence, a minor temperature change is necessary to degrade the carotenoids faster. Furthermore, the positive $\Delta H^\ddagger$ values (Table 1) for all solvents showed that the processes were endothermic, and external energy was required to degrade the carotenoids from the EUP samples. Additionally, higher $\Delta H^\ddagger$ values were observed for the extracts obtained with the mixture AC:ET and DES, revealing that these solvents provided a more susceptible media for the degradation of carotenoids with the increase of temperature. Concerning $\Delta S^\ddagger$ values (Table 1), all values were negative, indicating that the processes were less disordered. Finally, the $\Delta G^\ddagger$ values were all positive, showing the non-spontaneously of the systems (Table 1).

Overall, the thermal degradation assays suggest that the DES-based solvents, particularly those composed of octanoic acid, are the best choice for preserving carotenoids recovered from EUP at 40 °C and 50 °C. Nevertheless, at 60 °C, SUPRAS_{Oct} phase provided the most suitable environment for stabilizing the target bioactive compounds. The conventional AC:ET mixtures and ILs were found to have low thermal stability. This section reveals that the choice of the solvent for the recovery of carotenoids from vegetal biomass should consider its ability to preserve the biomolecules.

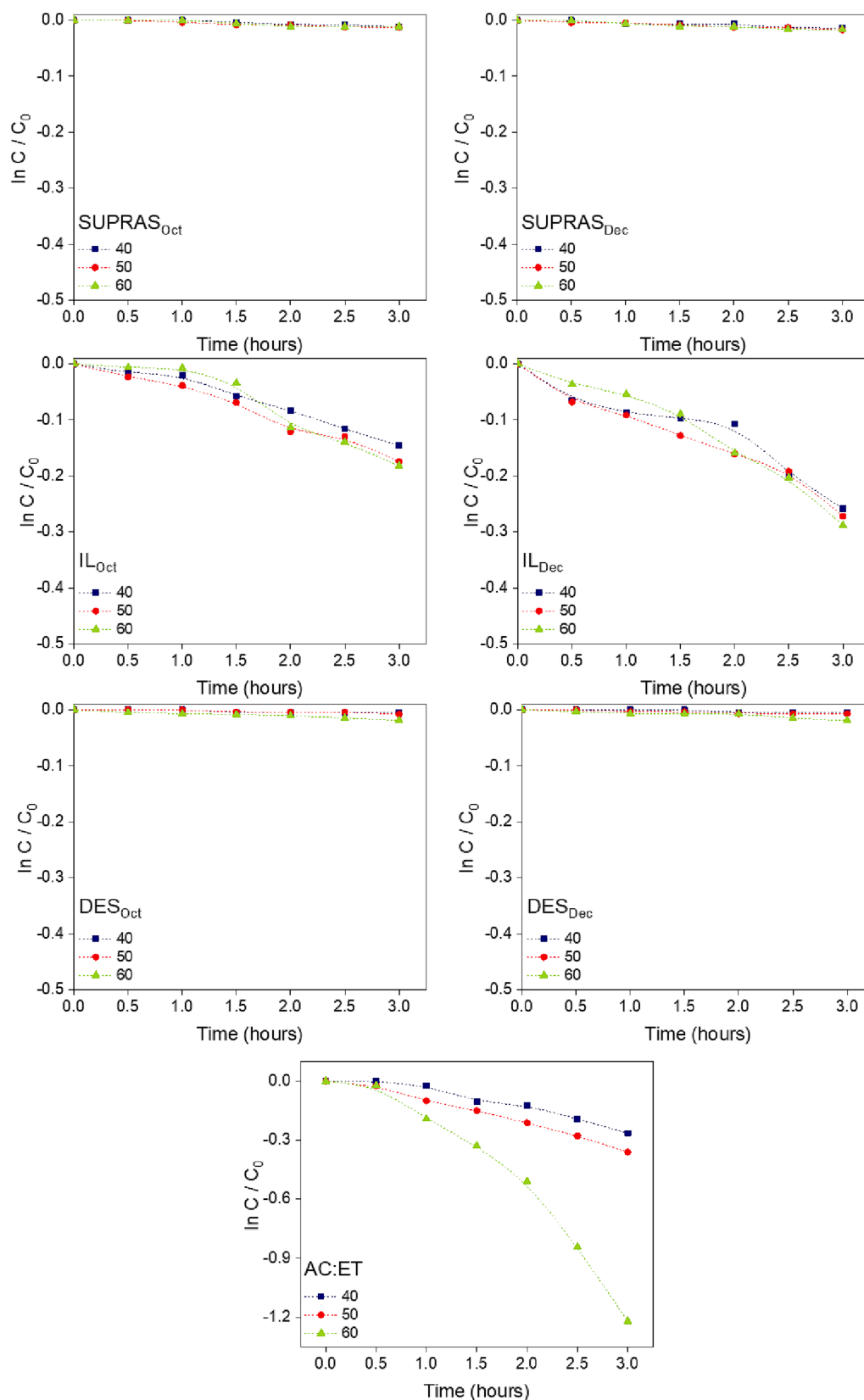


Fig. 5. Thermal stability of carotenoid compounds from EUP samples obtained with ILs and DES solutions at 45 °C and with SUPRAS and AC: ET at room temperature. The thermal degradation was evaluated at 40 °C, 50 °C, and 60 °C in the function of storage time (min). *The values were obtained from three replicates $\pm$ standard deviation (SD). Oct, octanoic acid. Dec, decanoic acid. AC:ET, a mixture of acetone and ethanol 1:1 (w/w).

Table 1

Thermodynamic parameters of carotenoids from EUP samples obtained with SUPRAS phase, AC:ET, and ILs and DES solutions during degradation.

Solvent	E_a (kJ/mol) (R^2)	$\Delta H^\ddagger$ (kJ/mol)	$\Delta S^\ddagger$ (kJ/mol)	$\Delta G^\ddagger$ (kJ/mol)			R^2
				40 °C	50 °C	60 °C	
SUPRAS _{Oct}	6.12 (0.96)	8.81	-0.17	61.24	62.92	64.59	0.99
SUPRAS _{Dec}	11.52 (0.98)	14.20	-0.15	60.91	62.40	63.89	0.98
DES _{Oct}	45.92 (0.89)	48.60	-0.05	63.51	63.99	64.46	0.90
DES _{Dec}	45.24 (0.90)	47.92	-0.05	63.38	63.88	64.37	0.91
IL _{Oct}	11.73 (0.98)	14.41	-0.13	54.80	56.09	57.38	0.99
IL _{Dec}	9.06 (0.88)	11.75	-0.13	53.78	55.12	56.46	0.97
AC:ET	63.75 (0.96)	66.43	0.04	53.65	53.25	52.84	0.89

3.2.2. Light degradation

The stability of carotenoids recovered from EUP was also evaluated through degradation assays using UV and white light, as illustrated in Fig. 6A and B, respectively. The coefficients of determination (R^2), half-life time ($t_{1/2}$), and k values are detailed in Tables S4 and S5 from the Supplementary Material.

The results depicted in Fig. 6A and B, revealed that all carotenoid-rich extracts are susceptible to degradation when exposed to UV light, with the IL-based solvents demonstrating the highest protective capacity of the target compounds (carotenoids). Carotenoids are highly reactive compounds that can undergo changes due to degradation and *cis-trans* isomerization, which affect their color and biological activity [23]. Thus, it was possible to observe that exposure to UV light was responsible for the color degradation of carotenoids from EUP. This phenomenon is believed to be related to the formation of singlet oxygen species in the presence of UV-light [24]. Under light conditions, carotenoids can be converted to the excited triplet state due to the absorption of high-energy light. Consequently, the formation of singlet oxygen is observed, which binds to the hydrocarbon chains of carotenoids, leading to the breakdown and/or degradation of these pigments [25]. Similar results were observed in studies developed by Spada et al. [26] and Lombardelli et al. [27], in which the authors found a reduction in the stability of carotenoids in the presence of UV light.

Regarding the protective effect promoted by the solvent used, it was observed that the IL, composed of decanoic acid, had a higher $t_{1/2}$ of

216.6 h (data available in Supplementary Material, cf. Table S2) compared to the other solvents evaluated. The solvent with the lowest capacity to stabilize carotenoids was the AC:ET mixture ($t_{1/2} = 17.1$ h), followed by SUPRAS based on octanoic acid ($t_{1/2} = 20.1$ h). Furthermore, in the SLE conducted with SUPRAS, it was observed that both for UV light degradation (Fig. 6A) and LED light (Fig. 6B), the increase in the alkyl chain length intensified the stability of the target carotenoids. This fact is strongly related to enhancing the hydrophobic regions formation in the solvent, which provides better protection of the carotenoids against the photooxidation process. Moreover, the water presence in formulated SUPRAS may have favored the degradation of carotenoids due to oxygen in the reaction medium. The exposure of extracts to LED light (Fig. 6B) showed that ILs also exerted a stronger protective effect on carotenoids, as observed for the UV light effect.

In general, the results indicated that all carotenoid-rich extracts were susceptible to degradation when exposed to UV and LED light, and ILs solutions demonstrated the highest protective capacity for carotenoids from EUP. Furthermore, the increase in alkyl chain length improved the preservation capacity of the solvent. The study highlights the importance of selecting an appropriate solvent for the preservation of carotenoids, as the effect of external factors such as temperature and light on the stability of carotenoids were found to be dependent on the solvent used. This emphasizes the need for a thorough understanding of the interaction between the solvent and carotenoids to develop effective preservation strategies. Further research is needed to explore the underlying mechanisms of carotenoid degradation and preservation in greater detail.

3.3. Recovery of carotenoids and recycle of SUPRAS phase and ILs solutions

Taking a sustainable approach to industrial processing and given the instability of carotenoids recovered from EUP, it is critical to integrate a process for carotenoid recovery and solvent recycling. This study proposes an integrated platform for carotenoid recovery from EUP biomasses, with a future goal of implementing circular and sustainable principles on an industrial scale. Following SLE using SUPRAS phase and ILs solution composed of octanoic acid (solvents with the highest recovery yield), at room temperature and at 45 °C for 60 min, respectively, the resulting-colored solutions containing carotenoids and solvents undergo a further process, as demonstrated in Fig. 7A and B. For IL-based extracts, ethyl acetate was added as an antisolvent to create a biphasic system. Concerning SUPRAS phase-based extract, the pH was adjusted to recover the target compounds and the octanoic acid by back-extraction.

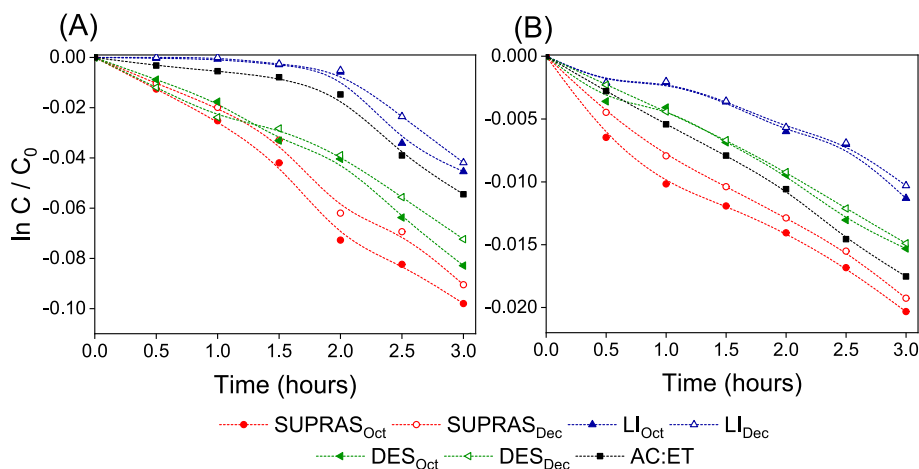


Fig. 6. (A) UV-Light and (B) LED-Light stability of carotenoid compounds from EUP samples obtained with ILs and DES solutions at 45 °C and with SUPRAS and AC:ET at room temperature. The light degradation was evaluated at 25 °C in the function of storage time (hours). *The values were obtained from three replicates $\pm$ standard deviation (SD). Oct, octanoic acid. Dec, decanoic acid. AC:ET, mixture of acetone and ethanol 1:1 (w/w).

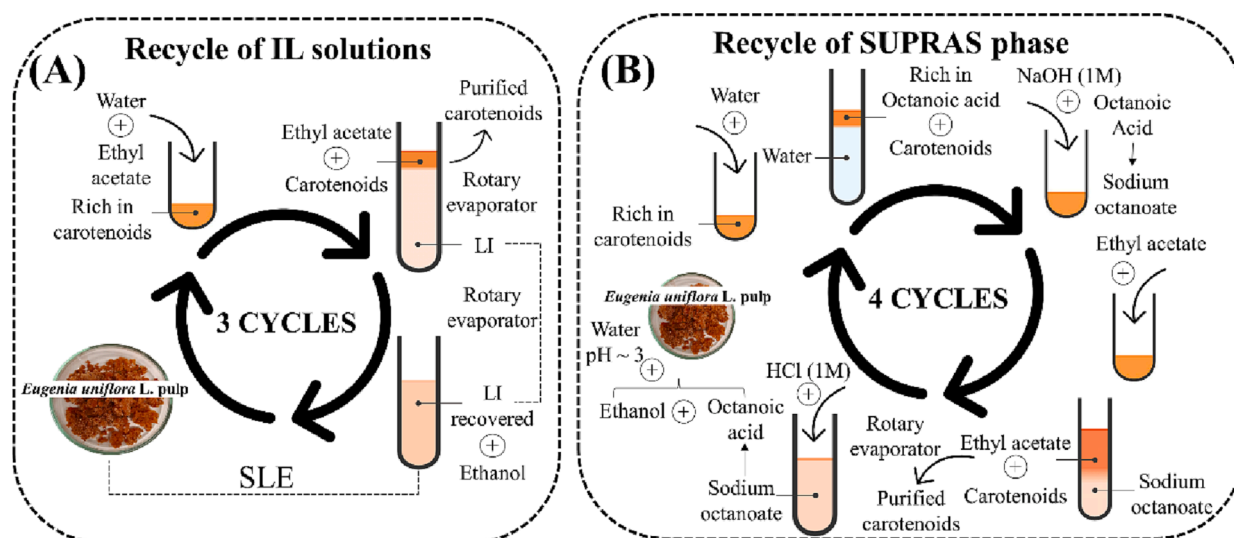


Fig. 7. Recovery of carotenoids from EUP and recycling processes of (A) ILs solution and (B) SUPRAS phase.

Besides, the selection of the solvent in the back extraction process was based on the ability to dissolve the carotenoids in volatile solvents, which was evaluated by the logarithm of the octanol–water partition coefficient ($\log K_{OW}$), as shown in Figure S3 (data available in the Supplementary Material). Thus, ethyl acetate was selected and previously evaluated for its ability to solubilize carotenoids from EUP samples. The SLE conditions evaluated were the same as for conventional extraction (1:40 w/v, 25 °C, and 60 min), exhibiting a carotenoids recovery of 10.23 ± 0.58 % of EUP samples.

The carotenoid recovery processes using recycled SUPRAS phase and ILs solution were conducted, and the yields (%) of carotenoids were compared with those obtained with fresh solvents, as shown in Fig. 8. Hence, four SLE cycles were evaluated to demonstrate the recycling process's feasibility. The first cycle used fresh SUPRAS phase and ILs solution, while in the following four cycles, the recycled SUPRAS and ILs solution was reused with ethyl acetate.

According to Fig. 8, the SUPRAS phase can be reused up to four

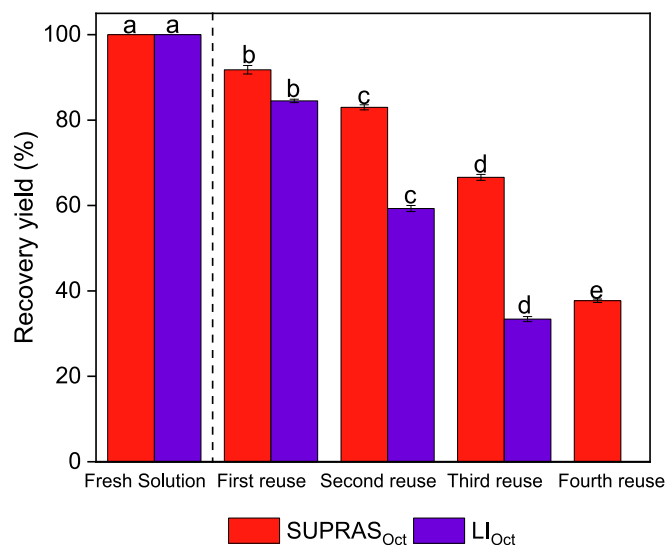


Fig. 8. Recovery yield (%) of carotenoid compounds from EUP samples using fresh and recycled ILs solution and SUPRAS phase, composed of octanoic acid as the solvents. *The values were obtained from three replicates $\pm$ standard deviation (SD). Different superscript letter for the same color of column means significant difference according to ANOVA and Tukey test ($p < 0.05$). Oct, octanoic acid.

times, while the ILs solution can be recycled three times without needing additional fresh solvent (to make up). The recovery yields of carotenoids remain stable (with a decrease in recovery yields lower than 20% in the first two recycles). However, after the third recycling cycle, the recovery of carotenoids with the SUPRAS phase is about 60%, with a decrease of approximately 40% compared to fresh solvent. For the ILs the recovery is about 30%, with a decrease of approximately 70% compared to the fresh solution. Although both solvents seem promising alternatives for recovering carotenoids from vegetable matrices, implementing a circular and sustainable industrial extraction platform using the SUPRAS phase and ILs solvent recycling presents significant challenges, primarily due to the low vapor pressure of these compounds. However, adding a back-extraction unit for solvent recycling and carotenoid recovery using bio-based solvents such as ethyl acetate is always necessary to overcome these limitations.

In addition, the potential of each integrated process for carotenoid recovery and solvent reusability was also investigated concerning the economic viability. Therefore, total costs were estimated based on the chemicals and electricity required in all stages (*i.e.*, production, extraction, purification, and polishing). As shown in Figure S5 (data available in the Supplementary Material), the SUPRAS phase-based process (USD 11.87) exhibited a lower cost than the IL-based (USD 28.53). This difference is mainly related to the IL production stage, which requires a long period of time for this solvent synthesis (72 h).

4. Conclusions

This work proposes the recovery of carotenoid compounds from EUP by utilizing ILs, DES, and SUPRAS, which are composed of carboxylic acids and serve as green and designer solvents. The results revealed that the octanoic acid-based SUPRAS phase was the most effective solvent for the extraction of natural carotenoids from EUP. This finding could be related to the simultaneous formation of the supramolecular network structure, offering a suitable microenvironment for the imprisonment of the hydrophobic target compounds. In addition to the extraction mechanism, a positive correlation was found between the SUPRAS phases', pH, and the carotenoids recovery, indicating that a more acidic environment favors the SLE process.

Regarding the thermal stability of the carotenoids, DES_{Oct} was the most effective solvent at 40 °C and 50 °C, while SUPRAS_{Oct} promoted the highest capacity to preserve the target compounds at 60 °C. Moreover, according to the E_a values, the SLE with AC: ET and DES as solvents were more susceptible to temperature enhancement. The thermodynamic

parameters showed that the degradation processes were endothermic and non-spontaneous. Related to the UV and LED-light stability, the ILs solutions provided a higher protective effect on the carotenoid compounds. Lastly, this work showed that the IL solution and SUPRAS phases could be employed in a circular platform for up to three and four cycles, respectively. Nonetheless, according to the economic viability investigation, the SUPRAS process exhibits a lower cost than the IL/Oct-based process.

Overall, it is worth mentioning that ILs, DES, and SUPRAS have some advantages and drawbacks of being employed, in which the choice of the solvent will depend on their future application and the process prerequisites. All solvents evaluated are tunable and environmentally friendly solutions, highlighting their capacity to extract carotenoids from EUP samples.

CRediT authorship contribution statement

Karina Mayumi Ueda: Conceptualization, Methodology, Investigation, Formal analysis, Data curation, Visualization, Writing – original draft. **Fernando Castro Leal:** Conceptualization, Investigation, Formal analysis, Data curation, Writing – review & editing. **Fabiane Oliveira Farias:** Conceptualization, Methodology, Investigation, Supervision, Writing – review & editing. **Cassamo U. Mussagy:** Methodology, Investigation, Data curation, Writing – original draft. **Luciana Igarashi-Mafra:** Resources, Funding acquisition, Writing – review & editing. **Marcos R. Mafra:** Conceptualization, Project administration, Resources, Supervision, Visualization, Funding acquisition, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.seppur.2023.124632>.

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